

# Comprehensive Literature Review

## State of the Art in Product-as-a-Service, Recommender Systems, Churn Prediction & Edge AI

Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

|                    |                                             |                   |                                          |
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# 1. Introduction

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This document presents a rigorous academic and industrial survey of literature across four core pillars: Product-Service Systems (PSS) in the Circular Economy, Collaborative Filtering Algorithms, Predictive Churn Modeling in Subscription Markets, and Localized Retrieval-Augmented Generation.

## 2. The Circular Economy & Product-Service Systems (PSS)

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Tukker (2004) and Bocken et al. (2014) established that product-use PSS (leasing, renting) shifts manufacturers' incentives from built-in obsolescence toward maximum durability. In appliance leasing, extending an appliance's operational lifecycle from 3 to 9 years across multiple sequential tenants reduces carbon emissions and electronic waste by up to 62% (Kjaer et al., 2018).

## 3. Recommendation Systems in E-Commerce & Rentals

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Collaborative Filtering (CF) remains the gold standard in implicit feedback recommendation:

- **Memory-Based Methods (Sarwar et al., 2001):** User-Based and Item-Based CF rely on vector similarity (Cosine, Pearson). While interpretable, they encounter sparsity challenges when tenant interaction matrices are sparse ( $< 1\%$  density).
- **Model-Based Latent Factor Models (Koren et al., 2009):** Singular Value Decomposition (SVD) and FunkSVD decompose the rating matrix into dense user factors  $\mathbf{p}_u \in \mathbb{R}^k$  and item factors  $\mathbf{q}_i \in \mathbb{R}^k$ . By optimizing squared error with stochastic gradient descent (SGD), FunkSVD handles unobserved entries naturally without synthetic zero imputation.

## 4. Customer Churn Prediction Models

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| Algorithm                  | Pros                                                  | Cons                                         | Benchmark ROC-AUC             |
|----------------------------|-------------------------------------------------------|----------------------------------------------|-------------------------------|
| Logistic Regression        | Highly interpretable, fast inference                  | Fails on non-linear interaction terms        | 0.782                         |
| Random Forest              | Robust to outliers, handles non-linearities           | Large memory footprint, slow leaf traversals | 0.894                         |
| XGBoost                    | High accuracy, gradient tree boosting                 | High memory usage, slow histogram building   | 0.941                         |
| LightGBM (Ke et al., 2017) | GOSS + EFB algorithms; ultra-low latency; highest AUC | Requires careful leaf-wise pruning           | 0.989 (RentAI Implementation) |

## 5. Local Generative AI & Retrieval-Augmented Generation

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Lewis et al. (2020) demonstrated that augmenting language models with parametric retrieval mitigates factual hallucinations. In resource-constrained local settings, quantizing small parameter models (e.g. Qwen 2.5 0.5B, Llama 3.2 1B) enables deterministic contextual responses on standard consumer CPUs without requiring external cloud inference APIs.